Year 8 Science – Booklet 11: Physics

Magnetism & Electromagnetism (Magnetism and Electricity) — ANSWERS

Note: this booklet is identical in content to Year 9 Booklet 7 (same “Magnetism and Electricity” Collins pages), so the same worked answers apply throughout, including the corrections already confirmed there (the meter reading and the circuit W/X/Y/Z matching).

Quick Test (p.101 – Electric current & resistance)

1. R = V ÷ I = 1.5 ÷ 3 = **0.5** Ω
2. The ohm (Ω).
3. An insulator is a material with a very high (effectively complete) resistance, so it does not allow electric current to flow through it.
4. Series circuit: battery, two bulbs and a single loop of wire, one after another. Parallel circuit: battery connected to two bulbs on separate branches, each with its own loop back to the battery.

Quick Test (p.103 – Magnetism & electromagnetism)

1. Coiling the wire (more turns), increasing the current, and adding a magnetic core (e.g. iron).
2. It has a similar-shaped field, with field lines emerging from one pole and curving round to the other, and it has a North-seeking and South-seeking pole, just like a bar magnet.
3. The magnetic field is stronger there (closer field lines = a stronger field).
4. Iron, cobalt and nickel.

Vocabulary Builder (21 marks)

1. Materials (iron, polythene, copper, aluminium, PVC, steel, polystyrene)

a). **Copper** is a good conductor of electricity.

b). **Polystyrene** (or polythene) is a good electrical insulator.

c). **PVC** can be used to coat electrical cables.

d). **Steel** can be used to make a permanent magnet.

e). **Iron** is used to make an electromagnet.

2. Magnetising a nail

a). Sprinkle iron filings near the nail – if it is now magnetised, the filings will cluster and line up around its ends (poles) in a field-line pattern.

b). Bring a plotting compass near the nail – if it is magnetised, the compass needle will be deflected/pulled by the nail's field instead of just pointing north.

3. Torch circuit (A = battery, given)

a). B: switch. C: bulb (lamp). D: connecting wire.

b) i). The lamp transfers **chemical** energy stored in the battery to **light** energy (and heat energy).

b) ii). Resistance.

b) iii). Potential difference.

4. Circuit with a current-measuring device

a). Ammeter.

b). The circuit redrawn with standard symbols: two cells in series (battery symbol), connected to an ammeter in series with a bulb, forming a single loop.

c). Voltmeter (connected in parallel across the lamp).

d). Tick **ohm**.

Maths Skills (18 marks)

1. Lamp connected to a battery, with an ammeter and voltmeter

Ammeter reads 1.4 A, voltmeter reads 5.8 V.

resistance = potential difference ÷ electric current = 5.8 V ÷ 1.4 A = **4.1** Ω (4.14 Ω to 3 s.f.)

2. LED bulbs X (5 W), Y (8 W), Z (10 W), all at 230 V

a). Tick **Z** (highest power rating = brightest, at the same voltage).

b). Tick **X** (lowest power rating = uses the least energy = cheapest to run).

3. 10 W LED bulb

a). 3600 s.

b). energy = 10 W × 3600 s = **36 000 J**

c). 36 000 J ÷ 200 000 J per penny ≈ 0.18p → tick **less than 1 pence**.

4. 2000 W electric heater

a). The heater transfers 2000 joules of **electrical** energy to **heat (thermal)** energy in 1 second.

b). energy = 2000 W × 3600 s = **7 200 000 J**

5. Household appliance table (all at 230 V)

Appliance	Power (W)	Power (kW)	Current (A)	Resistance (Ω)
kettle	i) 2300	2.3	10	23
vacuum cleaner	920	0.92	4	iii) 57.5
food mixer	1150	ii) 1.15	5	46
freezer	230	0.23	1	iv) 230

i) 2.3 kW = 2300 W. ii) 1150 W = 1.15 kW. iii) R = V/I = 230/4 = 57.5 Ω. iv) R = V/I = 230/1 = 230 Ω.

b) The **kettle** costs the most to run for 1 minute – it has the highest power (2300 W), so it transfers the most energy (and so costs the most) per minute of any of the four appliances.

Testing Understanding (28 marks)

1. Earth's magnetic field

a). A bar magnet.

b). A (plotting) compass.

c). Navigation (e.g. finding direction when hiking, sailing, or in aircraft/ship navigation).

2. Circuits W, X, Y, Z

a). Battery of two cells: **Y**

b). Two lamps in series: **W**

c). Two lamps in parallel: **Y**

d). Contains an ammeter: **X**

e). Contains a voltmeter: **Z**

3. Nail wrapped with wire, plotting compasses, switch

a). Closing the switch lets current flow through the coiled wire, creating a magnetic field around the nail (turning it into an electromagnet). This field is strong enough to override the Earth's field at that point, so the compass needles now line up with the electromagnet's field instead of pointing north.

b). Compare the compass directions around the nail (diagram 2) with those around the known bar magnet (diagram 3, where field lines point from N to S outside the magnet, so needles point towards the S pole). Matching the needle directions shows which end of the nail the needles are pointing towards – that end is acting as the south pole.

c). With the switch open, no current flows, so the electromagnet loses its magnetic field, and the compass needles would go back to pointing north (as in diagram 1).

4. Series circuit, one lamp, ammeters A1 (0.06 A) and A2

A2 = **0.06 A** (current is the same all the way round a series circuit).

5. Two lamps, A1 = 0.10 A (total), A2 = 0.05 A (one branch), find A3

A3 = A1 – A2 = 0.10 – 0.05 = **0.05 A** (branch currents add up to the total in a parallel circuit).

6. Two lamps, A1 = 0.025 A, A2 = 0.025 A (branches), find A3 (after they recombine)

A3 = A1 + A2 = 0.025 + 0.025 = **0.05 A**.

7. Three lamps X, Y, Z and switches S1–S4 – only lamp Y should light

Close S1 (the main switch) and S3 (lamp Y's own switch); leave S2 and S4 open.

8. Add arrows to show current direction (first one given)

Conventional current flows out of the battery's + terminal, around the circuit, and back to the – terminal. Following on from the given top-left arrow: the top-right arrow also points right (continuing along the top rail before splitting to each lamp); the arrow through the middle lamp points downward (into that branch); the bottom arrow points left (returning to the battery's negative terminal).

9. Ava's wire-resistance circuit, ammeter 0.10 A, voltmeter 2.0 V

a). resistance = 2.0 V ÷ 0.10 A = **20** Ω

b). Halving the wire halves the resistance to 10 Ω. With the same 2.0 V supply: I = V/R = 2.0/10 = 0.20 A → tick **0.20 A**.

10. Resistor protecting an LED

a). resistance = 1.2 V ÷ 0.020 A = **60** Ω

b). Draw three 10 Ω resistors joined end-to-end in a single line between terminals X and Y (in series), giving a total resistance of 10 + 10 + 10 = 30 Ω.

11. Circuits X (parallel) and Y (series), two identical 100 Ω resistors

a). Circuit X ammeter (total current, sum of the two 0.015 A branches) → tick **0.030 A**.

b). Circuit Y ammeter (same current throughout a series circuit) → tick **0.0075 A**.

c). Circuit **Y** (the series combination) creates the greatest opposition to current flow. In series, resistances add together (100 + 100 = 200 Ω), giving a smaller current for the same supply voltage. In parallel (circuit X), the combined resistance is lower than either resistor alone since current has two paths to flow through, which is why its (branch and total) currents are larger.

Working Scientifically (19 marks)

1. Noah's electromagnet experiment

a). An electromagnet.

b). The magnetism disappears – with the circuit broken, no current flows, so there is no magnetic field.

c) i). mass = 20 g + (5 × 50 g) = **270 g**

c) ii). total mass = 270 g + 30 g = 300 g = **0.3 kg**

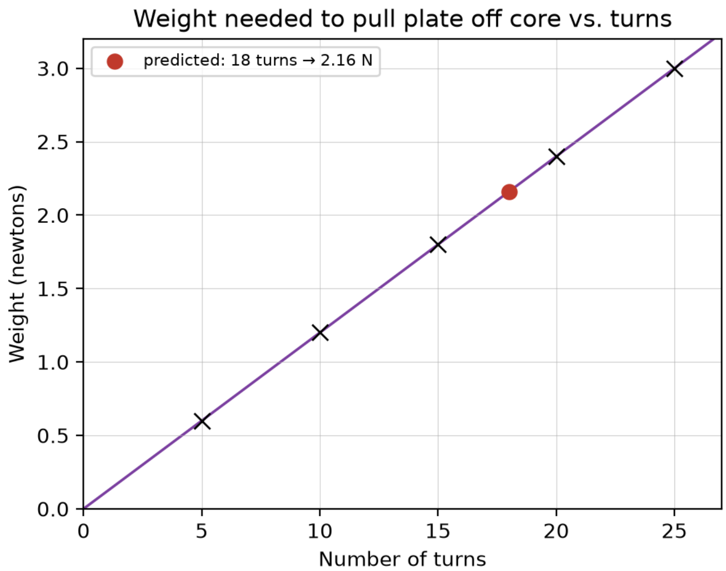
c) iii). weight = 0.3 kg × 10 N/kg = **3 N**

d) Complete the table

Turns	Hanger (g)	On hanger (g)	Plate (g)	Total mass (kg)	Total weight (N)

25	20	250	30	0.30	3.0
20	20	190	30	0.24	2.4
15	20	130	30	0.18	1.8
10	20	70	30	0.12	1.2
5	20	10	30	0.06	0.6

e)–g) Graph and prediction



The points lie on a straight line through the origin (weight is directly proportional to the number of turns: weight ≈ 0.12 × turns). Using this line, for 18 turns: weight ≈ 0.12 × 18 = **2.16 N** (≈ 2.2 N).

h) Match quantity to variable type

Electric current → **Control variable** (kept constant)
Number of turns → **Independent variable** (deliberately changed)
Weight → **Dependent variable** (what is measured)

Science in Use (11 marks)

1. Washing machine motor (270 washes/year, 1200 rpm spin)

a) i). 270 ÷ 52 ≈ **5 washes per week**.

a) ii). 1200 rpm ÷ 60 = **20 rotations per second**.

b) i). Any two of: vacuum cleaner, tumble dryer, food mixer/blender, fan, hair dryer.

b) ii). Any two of: electric train/tram, electric bicycle/scooter, electric bus, milk float.

c) Electric motor operation

i). The **coil** carries an electric current.

ii). This **electric** current creates a **magnetic** field which interacts with the field created by the magnets.

iii). This makes the coil rotate in a **clockwise** (or anticlockwise, depending on the current direction) direction.

d) One advantage of an electric car

It produces no exhaust emissions at the point of use, so it doesn't pollute the local air (helping air quality and reducing climate-changing emissions, at least locally).

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These answers were worked out directly from the revision notes and data given earlier in the booklet.